PROJECT PROPOSAL | AI CAPSTONE (MODULE 27)

# ResuMatch AI

Intelligent Resume - JD Gap Analysis & Career Coaching System

## 01 PROJECT OVERVIEW

| | |
|---|---|
| **Project Name** | ResuMatch AI |
| **Student Name** | Md. Shafat Hossain |
| **Mentor** | Sazzad Hossain |
| **Submission Date** | October 12, 2026 |
| **Target Objective** | A real-world, AI-driven web application that optimizes resumes against Job Descriptions (JDs) using Semantic Vector Embeddings and Generative AI. |

## 02 PROBLEM STATEMENT & REAL-WORLD IMPACT

- **Traditional ATS inefficiency.** Rigid keyword filtering rejects qualified candidates who word a skill differently (e.g., "data visualization" vs. "dashboarding"), rewards keyword stuffing, and gives no explanation for rejection.
- **Candidate skill gaps.** Job seekers rarely know how closely a resume matches a specific JD, which skills are missing, or which strengths are under-represented.
- **No real-time feedback.** Existing tools are generic checklists or paid services; they lack role-specific bullet-point rewriting, explainable scoring, and progress tracking across tailored versions.
- **Impact.** ResuMatch AI gives every candidate, especially fresh graduates and career-switchers, a measurable match score, a precise gap report, and actionable rewrite guidance at no cost.

## 03 PROPOSED SOLUTION & CORE FEATURES

1. **PDF Resume Parsing & Semantic Matching.** Extracts resume text, embeds resume and JD, and computes cosine similarity for an explainable 0-100 score based on meaning, not exact keywords.
2. **AI Skill & Keyword Gap Analysis.** Gemini returns Pydantic-validated structured JSON: matched skills, missing hard/soft skills, absent keywords, and gaps ranked by importance.
3. **Interactive AI Rewrite Coach.** Select any bullet and get JD-aligned rewrites with action verbs, quantified impact, and relevant keywords, without fabricating experience. Accept, refine, regenerate, and re-score instantly.
4. **Performance & History Analytics Dashboard.** Stores every analysis per user and visualizes score trends over time, recurring skill gaps, and before/after improvement across versions.

## 04 TECHNICAL ARCHITECTURE & TECH STACK

| Layer | Technology |
|---|---|
| **Frontend** | Next.js 14 (App Router), TypeScript, Tailwind CSS, Shadcn UI, Recharts |
| **Backend** | Python (FastAPI), PyPDF2, Pydantic |
| **AI/ML Layer** | Google Gemini API (Flash) for structured JSON output and vector embeddings |
| **Database** | Supabase (PostgreSQL with pgvector) |
| **Deployment** | Vercel (Frontend), Render (Backend) |

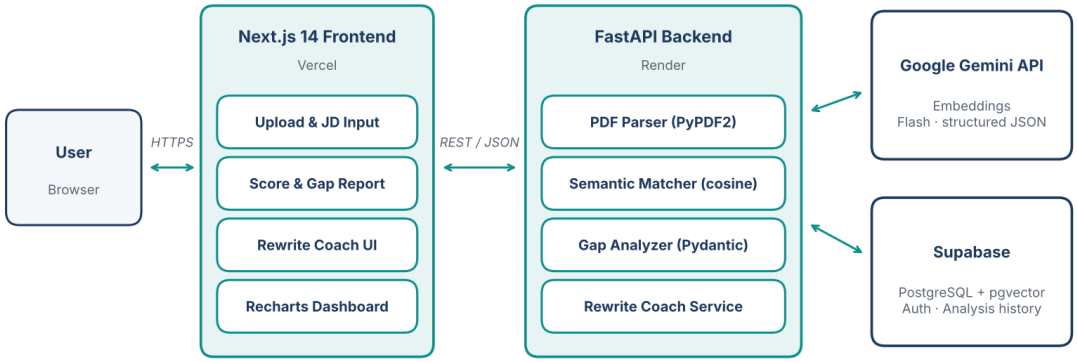



*Figure 1. System architecture and request flow*

- **Typed contracts end-to-end:** Pydantic schemas and TypeScript types for reliable LLM output.
- **pgvector storage:** efficient similarity queries and a base for future job recommendations.

## 05 DEVELOPMENT ROADMAP & MILESTONES

| Phase | Focus | Key Tasks | Outcome |
|---|---|---|---|
| **Phase 1** | Architecture Setup | GitHub repo and branching; scaffold Next.js and FastAPI; Supabase schema; environment config | Running skeleton with DB |
| **Phase 2** | AI Core & API | PDF pipeline; embeddings and cosine scoring; Gemini gap-analysis prompts; rewrite endpoint; validation | Functional AI API |
| **Phase 3** | Interactive Dashboard | Upload and JD UI; score and gap views; rewrite coach UI; Recharts history dashboard; responsive design | Complete user-facing app |
| **Phase 4** | Deployment & Testing | Deploy to Vercel and Render; unit and integration tests; scoring sanity checks; README and final report | Live, documented product |

## 06 EXPECTED DELIVERABLES & SUCCESS CRITERIA

| Deliverable | Contents | Acceptance Criteria |
|---|---|---|
| **Working Web Application** | Resume upload, match score, gap report, rewrite coach, history dashboard | Live public URL; end-to-end flow works without errors |
| **GitHub Repository** | Frontend, backend, and AI code with branching and meaningful commits | Reproducible local setup from README |
| **Documentation** | README (setup, usage, screenshots), AI methodology, Final Report (3-5 pages) | Covers design decisions, AI workflow, and results |

### Success Metrics

- **Score validity:** matched resume-JD pairs consistently outscore mismatched pairs on a curated sample set.
- **Responsiveness:** full analysis returned within ~10 seconds for a typical 2-page resume.
- **Reliability:** at least 95% of LLM responses pass schema validation (with automatic retry).

### Risks & Mitigation

| Risk | Mitigation |
|---|---|
| **API rate limits / quotas** | Cache embeddings, retry with exponential backoff, request throttling |
| **Scanned or unreadable PDFs** | Detect empty extraction and return a clear, actionable error |
| **Hallucinated rewrite suggestions** | Constrain prompts against fabrication; show original beside suggestion |
| **Free-tier cold starts** | Loading states and a health-check endpoint |
| **Resume data privacy** | Per-user row-level security in Supabase; user-initiated deletion |

*Submitted by Md. Shafat Hossain for review by Mentor Sazzad Hossain*